

AMIT MANJARLY

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TECHNICAL SKILLS

ML & Deep Learning: CNNs, ResNet, U-Net, Open-Set Recognition, Causal ML, Conformal Prediction, FNO / PINNs

LLMs & GenAI: Llama-3.2, Mistral, RAG, Fine-Tuning (QLoRA, rsLoRA), Hugging Face, ChromaDB

Computer Vision: YOLOv8/v11, OOD Detection, TorchGeo, Object Detection, Image Segmentation, CV Models

ML Frameworks: PyTorch, PyTorch Lightning, TensorFlow, Scikit-learn, XGBoost, LightGBM, NumPy, Pandas

Infra & MLOps: Docker, Kubernetes, MLflow, DVC, CUDA, Apple MPS, FastAPI, Streamlit, Google Earth Engine

Languages & DBs: Python (Advanced), SQL, C++; PostgreSQL, InsForge, SQLite, Hive, Impala, Oracle

EXPERIENCE

Resilient

Jul 2026 - Present

Machine Learning Engineer Intern – Tempe, AZ

- Built cocoa digital twin using Sentinel-1/2 and rsLoRA fine-tuning, reducing model weights 44.3x to achieve 0.91 F1.
- Engineered 2D Fourier Neural Operator surrogates and climate MoE models, yielding up to 0.927 F1 and 0.85 mIoU scores.
- Trained a 730-day Arabica coffee yield model and DynACof emulator, attaining a low 0.1418 t/ha validation RMSE metric.
- Constructed high-resolution agroforestry segmentation pipelines using TorchGeo, securing 0.83 mIoU and 0.90 F1 score.
- Applied causal Deep R-Learners and conformal prediction to estimate effects, achieving 90% interval coverage rate.

Accenture PLC

Mar 2021 - Jul 2024

Software Engineer – Hyderabad, India

- Built Python automated testing frameworks for enterprise software, decreasing manual QA pipeline cycles by 40%.
- Optimized high-volume SQL integration pipelines processing millions of rows, decreasing data delay times by 33%.
- Refactored ETL workflows across Oracle and Hive data systems to boost pipeline throughput and runtime reliability.
- Automated validation rules and quality checks in Python, improving operational reliability of downstream pipelines.
- Profiled system workloads to remove critical bottlenecks, driving a 25% efficiency improvement in execution loops.
- Constructed production monitoring dashboards with analytics engineers to track system telemetry and data health.
- Led technical onboarding and knowledge-transfer workshops, standardizing ETL development workflows across teams.

PROJECTS

Open-World Continual Learning for Autonomous Agents [\[github\]](#)

Jan 2026 - Apr 2026

Product Optimization · Python · Architecture from Scratch · End-to-End Deployment

- Constructed continual perception pipelines with YOLO detection and domain adaptation across Waymo and nuScenes sets.
- Implemented Elastic Weight Consolidation using diagonal Fisher information to prevent catastrophic model forgetting.
- Formulated entropy, maximum-softmax, and energy-based OOD classification layers to flag previously unseen obstacles.
- Engineered FastAPI inference service and Streamlit dashboard containerized via Docker and tracked using MLflow/DVC.
- Implemented automated unit tests for batch loading, continual-learning modules, and uncertainty estimation suites.

LLM Fine-Tuning for Mathematical Reasoning (STaR) [\[github\]](#)

Sep 2025 - Oct 2025

Large Language Models · Deep Learning · Strategic Roadmap · Iterative Improvement

- Evaluated Zero-Shot CoT, standard SFT, and STaR-style reasoning architectures on the standard GSM8K benchmark suite.
- Implemented parameter-efficient QLoRA with 4-bit quantization, decreasing target GPU memory requirements by 75%.
- Engineered bootstrap pipelines to generate and filter multi-step rationale trajectories prior to fine-tuning runs.

End-to-End Hybrid Recommendation Engine [\[github\]](#)

May 2025 - Jun 2025

Recommendation Systems · Predictive Modeling · Advanced Statistics · Metrics

- Built a hybrid recommender merging collaborative and content-based filtering algorithms on the MovieLens 100K data.
- Achieved Precision@10 score of 0.26 against a 0.15 baseline reference, demonstrating an overall 73% precision gain.
- Designed offline validation workflows to evaluate rank quality across models while resolving cold-start anomalies.

EDUCATION

M.S. in Data Science, Analytics and Engineering

Aug 2024 - May 2026

Arizona State University – Tempe, AZ

Coursework: Machine Learning, Deep Learning, NLP, Statistical Inference

B.E. in Computer Engineering

Aug 2015 - Jun 2019

Gujarat Technological University – Ahmedabad, India

Coursework: DSA, OS, DBMS, Computer Networks, OOP